Cessna 402C (C402C)

2× Continental TSIO-520-VB turbocharged, 325 hp · 9–10 (commuter) seats · Typical 402C (1979–1985, 6,850 lb) POH procedures; unpressurised, wet-wing fuel without tip tanks, hydraulic landing gear with emergency hand pump, airstair cabin door.

V-speeds & limits

Vmca	**80 KIAS (red radial)**
Vsse	**95 KIAS (intentional one-engine-inop)**
Vr	**approx 95 KIAS (liftoff, normal takeoff)**
Vx	**93 KIAS**
Vxse	**100 KIAS**
Vyse	**101 KIAS (blue line)**
Vy	**107 KIAS**
Vfe	**180 KIAS (15°) / 148 KIAS (full)**
Vle / Vlo	**180 KIAS extended and extension / 148 KIAS retraction (verify in POH)**
Va	**150 KIAS (6,850 lb)**
Vmo / Vne	**230 KIAS**
Vs / Vso	**approx 85 / 76 KIAS (max weight; verify)**
Vref	**approx 95 KIAS final (full flaps, max landing weight)**

Normal procedures

Before Start

Preflight inspection	**COMPLETE (fuel drains, wing lockers, nose baggage doors latched, induction / exhaust clear)**
Passenger briefing	**COMPLETE – commuter brief: belts, airstair door and emergency exits, no smoking, baggage stowed, briefing cards, electronic devices**
Seats, belts, shoulder harnesses	**ADJUST & LOCK (all passengers belted)**
Airstair cabin door	**CLOSED & LATCHED – upper and lower halves, handle flush, door light OUT (if installed)**
Parking brake	**SET**
Avionics power switch	**OFF**
Landing gear handle	**DOWN**
Throttles / propellers / mixtures	**IDENTIFY – throttles closed, props high RPM, mixtures idle cutoff**
Fuel selectors	**LEFT engine – LEFT tank, RIGHT engine – RIGHT tank (feel for detents)**
Fuel quantity	**CHECK both wings; LOW FUEL lights out**
Cowl flaps	**OPEN**
Alternate static source	**OFF (normal)**
Battery switch	**ON**
Alternator switches	**ON**
Landing gear lights	**THREE GREEN; unlocked light OUT**
Annunciator panel	**PRESS TO TEST**
Beacon	**ON**

Engine Start

Propeller area	**CLEAR (both sides)**
Magneto switches	**ON (both)**
Mixture	**FULL RICH**
Propeller	**HIGH RPM**
Throttle	**1/2 OPEN**
Auxiliary fuel pump	**HIGH until fuel flow peaks (3–5 sec), then OFF (LOW in hot weather)**
Throttle	**1/4 OPEN**

▶ Engine hot or flooded?

No:

Yes:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Mixture	**ADVANCE to FULL RICH as engine fires, throttle to idle**
Starter switch	**ENGAGE (release when engine fires)**
Throttle	**800–1000 RPM (do not exceed 1200 RPM until oil pressure stable)**
Oil pressure	**10 psi within 30 sec, rising to 30–60 psi**
Second engine	**REPEAT start sequence**
Alternators	**CHECK both charging (ammeters / voltmeter approx 28 V, ALT lights out)**
Hydraulic pressure	**CHECK (if gauge installed)**
External power	**DISCONNECTED (if used)**
Avionics power switch	**ON**
Engine instruments	**CHECK both (fuel flow steady, manifold pressure near ambient)**

Before Taxi

Brakes	**CHECK on initial roll**
Flight instruments	**CHECK & SET (altimeters, HSI / DG, turn coordinator during turns)**
Avionics / GPS	**SET**
Autopilot / electric trim	**CHECK & OFF**
Trims (elevator, rudder, aileron)	**SET for takeoff**
Fuel selectors	**CHECK LEFT / RIGHT (own tanks)**
Cowl flaps	**OPEN**

Before Takeoff

Parking brake	**SET**
Flight controls	**FREE & CORRECT**
Fuel selectors	**LEFT / RIGHT own tanks (not crossfeed) – feel for detents**
Mixtures	**FULL RICH (turbocharged – do not lean for takeoff at any elevation)**
Throttles	**1700 RPM (one engine at a time)**
Magnetos	**CHECK L & R each engine (150 RPM max drop, 50 RPM max difference)**
Propellers	**CYCLE 2–3 times (RPM drops, then returns)**
Propellers – feather check	**PULL to FEATHER momentarily at 1500–1700 RPM – RPM drops (approx 200–300 RPM); return before RPM falls below 1000**
Engine instruments	**CHECK both (oil pressure 30–60 psi, oil temp green, fuel flow steady, CHT / TIT green)**
Alternators	**CHECK both charging (load with pitot heat / lights)**
Gyro suction	**CHECK in green, both pump indicators clear**
Throttles	**IDLE check (600–650 RPM), then 1000 RPM**
Auxiliary fuel pumps	**LOW (HIGH only with a failed engine-driven pump)**
Cowl flaps	**OPEN**
Propellers	**HIGH RPM**
Flaps	**UP (15° short field) – check indicator**
Trims	**SET takeoff range**
Landing gear handle	**DOWN – IDENTIFY**
Pitot heat / de-ice	**AS REQUIRED**
Lights	**AS REQUIRED**
Transponder	**ALT**
Airstair door / emergency exits	**LATCHED, door light OUT**
Takeoff briefing	**COMPLETE (Vmca 80 / abort below 95 / Vyse 101; 39 in Hg max; engine-out plan)**
Takeoff power	**ADVANCE THROTTLES SLOWLY to 39 in Hg / 2700 RPM – avoid overboost; set power before brake release**

After Takeoff / Climb

Airspeed	**ACCELERATE to 101 KIAS (Vyse) minimum before gear retraction**
Landing gear	**UP (positive rate, no usable runway, below 148 KIAS)**
Gear lights	**OUT (gear up and locked)**
Flaps	**UP (if used)**
Climb power	**REDUCE SLOWLY to 32.5 in Hg / 2450 RPM cruise climb (max continuous per POH)**
Airspeed	**107 KIAS (Vy) / 120–130 KIAS cruise climb**
Mixtures	**FULL RICH or lean to climb fuel flow schedule; TIT below 1650°F**
Cowl flaps	**OPEN (CHT / oil temp in green)**
Auxiliary fuel pumps	**OFF (LOW if fuel flow fluctuates)**
Manifold pressure	**MONITOR – turbochargers hold climb power to critical altitude; re-adjust as required**
Engine instruments	**CHECK both**

Cruise

Cruise power	**SET SLOWLY (2300–2450 RPM, 26–32 in Hg typical; 60–75%)**
Mixtures	**LEAN per cruise table to approx 1550–1600°F TIT (never above 1650°F); enrich if rough**
Cowl flaps	**CLOSED (as required for CHT)**
Propellers	**SYNCHRONISE**
Fuel quantity / flow	**CHECK both wings against plan; balance with crossfeed only if required**
Fuel selectors	**LEFT / RIGHT own tanks (CROSSFEED only for one-engine-inop or imbalance in level flight)**
Engine instruments	**CHECK both (oil temp 240°F max, CHT 460°F max, TIT 1650°F max)**

Turbocharged engines: make all power changes slowly and smoothly. Set manifold pressure with the throttle, then RPM, then mixture. Momentary overboost above 39 in Hg must be reduced immediately and logged.

Altimeter / heading	**CHECK**

Descent

Altimeters	**SET**
Fuel selectors	**LEFT / RIGHT own tanks**
Mixtures	**ENRICH progressively**
Power	**REDUCE SLOWLY – approx 2 in Hg per minute, keep engines warm (avoid rapid cooling)**
Cowl flaps	**CLOSED**
Airspeed	**PLAN below 180 KIAS for approach flaps and gear**
Approach briefing	**COMPLETE (incl. single-engine go-around decision point)**
Passengers	**BRIEFED – belts fastened, seat backs upright, baggage stowed**

Before Landing

Fuel selectors	**LEFT / RIGHT own tanks**
Auxiliary fuel pumps	**LOW**
Mixtures	**FULL RICH**
Cowl flaps	**AS REQUIRED**
Airspeed	**BELOW 180 KIAS**
Landing gear	**DOWN – THREE GREEN, unlocked light out (identify gear handle)**
Flaps	**15° below 180 KIAS; FULL on final below 148 KIAS**
Propellers	**HIGH RPM (on final)**
Airspeed	**approx 95 KIAS final (full flaps)**
Landing lights	**ON**
Landing gear	**THREE GREEN – CONFIRMED on final**

After Landing

Flaps	**UP (identify flap handle before moving)**
Cowl flaps	**OPEN**
Auxiliary fuel pumps	**OFF**
Pitot heat / de-ice	**OFF**
Landing lights / strobes	**AS REQUIRED**
Transponder	**STBY / as required**
Trims	**SET takeoff**

Shutdown

Parking brake	**SET**
Throttles	**1000 RPM for 2–3 min – turbocharger cool-down (taxi time counts)**
Avionics power switch	**OFF**
Electrical equipment / lights	**OFF**
Mixtures	**IDLE CUTOFF**
Magneto switches	**OFF (after propellers stop)**
Battery & alternator switches	**OFF**
Control lock	**INSTALL**
Airstair door	**OPEN for passenger deplaning only after propellers stop**
Chocks / tiedowns	**AS REQUIRED**

Abnormal procedures

Alternator Failure / Low Voltage

ALT warning light, one ammeter zero or voltage below 24 V

Affected alternator switch	**OFF, then ON (reset over-voltage relay)**
Alternator field circuit breaker	**CHECK / RESET once**

▶ Alternator restored?

Yes:

Ammeters / voltmeter	**MONITOR both**

No – one alternator failed:

Affected alternator switch	**OFF**
Electrical load	**REDUCE to within remaining alternator**
Deferred: One alternator inoperative – electrical load reduced, land as soon as practical.	**NOTED**

No – both failed:

Both alternator switches	**OFF**
Non-essential electrical load	**REDUCE (battery only, limited endurance)**
Landing gear	**PLAN early extension while battery power remains for gear lights**
Deferred: Both alternators failed – battery only; extend gear early, flaps may be unavailable, land as soon as possible.	**NOTED**

Low Oil Pressure (one engine)

Oil pressure below 30 psi in flight

Oil pressure / oil temperature	**CHECK and cross-check the affected engine**

▶ Oil temperature rising or pressure below 10 psi?

Yes – genuine:

Affected engine power	**REDUCE to minimum required**
Engine	**SHUT DOWN and FEATHER before seizure if altitude and terrain allow (Engine Failure in Flight)**

Feather while oil pressure remains – the propeller and turbocharger controller are oil-operated.

Deferred: Low oil pressure – precautionary engine shutdown; single-engine landing, land as soon as possible.	**NOTED**

No – gauge suspect:

Flight	**CONTINUE, monitor oil temperature and pressure**
Deferred: Oil pressure indication suspect – maintenance check required.	**NOTED**

Turbocharger Failure / Manifold Pressure Loss

Manifold pressure falls toward ambient, power loss, possible exhaust smell

Affected engine throttle	**REDUCE to limit power on the unboosted engine**
Affected engine mixture	**ENRICH – fuel flow was scheduled for boosted power**
Affected engine TIT / CHT	**CHECK in limits**
Engine and nacelle	**CHECK for smoke, exhaust leak or fire indications**

▶ Smoke, fire indication or exhaust leak suspected?

Yes:

Affected engine	**SHUT DOWN and FEATHER (Engine Failure in Flight / Engine Fire)**
Deferred: Turbocharger / exhaust failure – engine secured, single-engine landing, land as soon as possible.	**NOTED**

No:

Affected engine	**CONTINUE at reduced (normally aspirated) power; expect loss of power with altitude**
Altitude	**DESCEND as required to maintain power and airspeed**
Deferred: Turbocharger failure – reduced power on one engine, land as soon as practical; exhaust system inspection required.	**NOTED**

High TIT / Overboost

TIT above 1650°F or manifold pressure above 39 in Hg

Affected engine throttle ..... **REDUCE SLOWLY until manifold pressure within limits**

Affected engine mixture ..... **ENRICH until TIT below 1650°F**

Affected engine cowl flap ..... **OPEN**

Airspeed ..... **INCREASE for cooling**

▶ TIT and manifold pressure back in limits?

Yes:

Power ..... **RESET smoothly, monitor**

Deferred: Overboost / high TIT occurred – log for maintenance inspection. ..... **NOTED**

No:

Affected engine power ..... **REDUCE further; shut down and feather if TIT cannot be controlled or power erratic**

Deferred: Turbocharger controller fault – reduced power on one engine, land as soon as practical. ..... **NOTED**

Fuel Flow Fluctuation / Vapour / Rough Running (one engine)

Affected engine auxiliary fuel pump ..... **LOW**

Affected engine mixture ..... **ADJUST for smooth running**

Fuel selector ..... **CHECK own tank – feel for detent; CROSSFEED if tank quantity suspect**

Affected engine magnetos ..... **CHECK L / R individually, then BOTH**

Engine instruments ..... **CHECK (CHT, TIT, fuel flow, oil pressure)**

▶ Fuel flow steady and engine smooth?

Yes:

Deferred: Fuel flow fluctuation cleared – leave aux pump LOW; report to maintenance. ..... **NOTED**

No – fuel flow very low (engine-driven pump failed):

Affected engine auxiliary fuel pump ..... **HIGH (reduces to low pressure automatically at low throttle settings)**

Affected engine mixture ..... **LEAN as required for smooth running (HIGH pump floods at low power)**

Deferred: Engine-driven fuel pump suspect – aux pump HIGH, land as soon as practical; expect rough running at low power. ..... **NOTED**

No – still rough:

Affected engine power ..... **REDUCE to smoothest setting – feather if power unusable (Engine Failure in Flight)**

Deferred: Engine rough – land as soon as practical at nearest suitable airport. ..... **NOTED**

Crossfeed Operation / Fuel Imbalance

One engine inoperative or wing tank imbalance

Flight attitude ..... **LEVEL (crossfeed in level flight only)**

Operating engine fuel selector ..... **CROSSFEED (opposite wing tank) – feel for detent**

Fuel flow ..... **CHECK steady after selection**

Auxiliary fuel pump (operating engine) ..... **LOW if fuel flow fluctuates**

Fuel quantity ..... **MONITOR – return to own tank before descent, approach and landing**

Deferred: Crossfeed used – selectors back to own tanks for landing. ..... **NOTED**

Propeller Overspeed

RPM above 2700 and not responding to prop lever

Affected engine throttle ..... **REDUCE**

Affected engine propeller ..... **LOW RPM (toward feather detent)**

Airspeed ..... **REDUCE to bring RPM within limits**

Affected engine oil pressure ..... **CHECK (governor is oil-operated)**

▶ RPM controllable at or below 2700?

Yes:

Power ..... **SET minimum that keeps RPM in limits**

Deferred: Propeller governor fault – control RPM with throttle and airspeed, land as soon as practical. ..... **NOTED**

No:

Affected engine ..... **SHUT DOWN and FEATHER (Engine Failure in Flight)**

Deferred: Engine shut down for prop overspeed – single-engine landing, land as soon as possible. ..... **NOTED**

Flap Failure / Asymmetry

Flap handle ..... **NEUTRAL / stop – do not continue extension**

Flap circuit breaker ..... **CHECK / RESET once**

▶ Flaps operable and symmetric?

Yes:

Flap setting ..... **AS REQUIRED**

No:

Aileron / rudder ..... **AS REQUIRED to hold wings level; airspeed above 105 KIAS**

Landing ..... **AT CURRENT FLAP SETTING – add 10 KIAS to approach speed (flaps up approx 105–110 KIAS), longer runway**

Deferred: Flaps inoperative / asymmetric – flaps-up or partial-flap landing, increased landing distance. ..... **NOTED**

Landing Gear Unsafe / Fails to Retract or Extend Normally

Red unlocked light, fewer than three green, or warning horn

Airspeed ..... **BELOW 148 KIAS**

Gear light bulbs / navigation light dimming ..... **CHECK (lights dim with nav lights on)**

Hydraulic pressure ..... **CHECK (if gauge installed)**

Landing gear handle ..... **CYCLE once to the desired position**

▶ Gear in desired position?

Yes:

Gear indication ..... **CONFIRM three green, unlocked light out**

Deferred: Gear system fault – do not retract after landing; maintenance inspection required. ..... **NOTED**

No – failed to retract:

Landing gear handle ..... **DOWN – leave down**

Airspeed ..... **BELOW 180 KIAS**

Deferred: Gear will not retract – gear down, reduced performance and range (negative single-engine climb), land as soon as practical. ..... **NOTED**

No – failed to extend:

Use Landing Gear Emergency Extension (Hand Pump).

Airstair / Nose Baggage Door Open in Flight

Aircraft ..... **CONTROLLABLE – fly the airplane**

Airspeed ..... **REDUCE to approx 120 KIAS (or less if buffet)**

Passengers ..... **SEATED, belts fastened, away from the door**

Do not attempt to close the airstair door in flight. An open nose baggage door can affect pitch and airflow to the engines – land as soon as possible, avoid abrupt manoeuvres and sideslips.

Deferred: Door unlatched – return and land, normal approach speed with slightly higher power; expect noise and buffet. ..... **NOTED**

Gyro Suction / Attitude Indicator Failure

Suction gauge ..... **CHECK (red pump indicator shows which pump failed – remaining pump maintains system)**

Attitude / heading indicators ..... **CROSS-CHECK with turn coordinator, compass, GPS, standby instruments**

Failed instruments ..... **COVER**

Autopilot ..... **OFF if coupled to failed gyro**

IMC ..... **EXIT / AVOID if both pumps or the attitude indicator failed**

Deferred: Gyro system failure – partial panel; prefer VMC approach. ..... **NOTED**

Inadvertent Icing

Pitot heat ..... **ON**

Propeller de-ice / windshield de-ice / surface de-ice boots ..... **ON (if installed and operative)**

Cabin heat / windshield defrost ..... **ON**

Alternate induction air ..... **AUTOMATIC – expect a manifold pressure drop; re-adjust throttles slowly**

Course / altitude ..... **CHANGE to leave icing conditions**

Power ..... **INCREASE to maintain airspeed (39 in Hg max)**

▶ Ice remaining on airframe for landing?

No:

Yes:

Approach ..... **FLAPS 15° or less, increase approach speed by 10–15 KIAS, no slips**

Deferred: Residual airframe ice – partial-flap landing, approach 10–15 KIAS faster, longer runway, higher stall speed. ..... **NOTED**

Pitot / Static Blockage

Erratic or frozen airspeed, altimeter or VSI

Pitot heat ..... **ON**

Alternate static source ..... **ON (pull) if static blockage suspected**

Cabin vents / heater ..... **CLOSED (alternate static reads cabin pressure)**

Airspeed ..... **FLY ATTITUDE and POWER; use GPS ground speed as a cross-check**

Deferred: Pitot-static fault – airspeed / altitude unreliable, fly known power and attitude on approach, allow altimeter error on alternate static. ..... **NOTED**

Emergency procedures

Engine Failure During Takeoff

On the runway or just airborne with runway remaining

▶ Airborne?

No – below approx 95 KIAS:

- Throttles (memory) — **CLOSED (both)**
- Brakes (memory) — **MAXIMUM – stop straight ahead**
- Mixtures — **IDLE CUTOFF**
- Fuel selectors — **OFF**
- Magneto switches — **OFF**
- Battery & alternator switches — **OFF**

Yes – runway remaining:

- Throttles (memory) — **CLOSED (both)**
- Land (memory) — **STRAIGHT AHEAD, gear down**
- Brakes — **MAXIMUM**
- Mixtures / fuel selectors / magnetos — **OFF after stop**

Yes – no runway remaining:

Go to Engine Failure After Liftoff.

Below Vmca (80 KIAS) directional control cannot be maintained with full power on one engine. Do not attempt to continue a takeoff below approx 95 KIAS.

Engine Failure After Liftoff

Airborne, insufficient runway to land ahead

LAND AS SOON AS POSSIBLE

- Mixtures (memory) — **FULL RICH**
- Propellers (memory) — **HIGH RPM**
- Throttles (memory) — **FULL (39 in Hg – do not overboost)**
- Flaps (memory) — **UP**
- Airspeed (memory) — **101 KIAS (Vyse); 100 KIAS (Vxse) only to clear obstacles – never below 80 KIAS**

▶ Landing gear position?

Gear DOWN:

With gear down and the propeller windmilling, single-engine climb is negative at most weights. Retract if committed to fly; otherwise land straight ahead.

- Landing gear (memory) — **UP (if committed to fly)**
- Inoperative engine (memory) — **IDENTIFY – dead foot, dead engine**
- Inoperative engine throttle (memory) — **VERIFY – retard, no change**
- Inoperative engine propeller (memory) — **FEATHER (before RPM falls below approx 800 – start locks)**
- Inoperative engine mixture (memory) — **IDLE CUTOFF**

Gear UP:

- Inoperative engine (memory) — **IDENTIFY – dead foot, dead engine**
- Inoperative engine throttle (memory) — **VERIFY – retard, no change**
- Inoperative engine propeller (memory) — **FEATHER (before RPM falls below approx 800 – start locks)**
- Inoperative engine mixture (memory) — **IDLE CUTOFF**

Bank — **3–5° toward operating engine, ball half out**

Trim — **RUDDER & AILERON for straight flight**

Climb — **101 KIAS to safe altitude, then Engine Securing**

Deferred: One engine inoperative after takeoff – single-engine approach and landing, no go-around once committed. — **NOTED**

Engine Failure in Flight

Loss of power on one engine at altitude

LAND AS SOON AS POSSIBLE

Mixtures (memory)	**FULL RICH**
Propellers (memory)	**HIGH RPM**
Throttles (memory)	**FULL (as required, 39 in Hg max)**
Landing gear / flaps (memory)	**UP**
Airspeed (memory)	**101 KIAS (Vyse) minimum**
Inoperative engine (memory)	**IDENTIFY – dead foot, dead engine**
Inoperative engine throttle (memory)	**VERIFY – retard, no change**
Inoperative engine propeller (memory)	**FEATHER**
Inoperative engine mixture (memory)	**IDLE CUTOFF**
Bank / rudder	**3–5° toward operating engine, ball half out**
Trim	**RUDDER & AILERON**

▶ Cause obvious (fuel) and altitude available for a restart?

Yes – attempt air start:

Use the Air Start procedure. If unsuccessful, secure the engine. A sudden manifold pressure drop with the engine still running is a turbocharger failure, not an engine failure – see Turbocharger Failure.

No – secure engine:

Continue with Engine Securing, then divert to the nearest suitable airport.

Deferred: One engine inoperative – single-engine approach and landing, drift-down altitude check, no go-around once committed. **NOTED**

Engine Securing (after feather)

Inoperative engine propeller feathered

Inoperative engine fuel selector	**OFF**
Inoperative engine auxiliary fuel pump	**OFF**
Inoperative engine magneto switches	**OFF**
Inoperative engine alternator switch	**OFF**
Inoperative engine cowl flap	**CLOSED**
Propeller synchrophaser	**OFF (if fitted)**
Electrical load	**REDUCE to within one alternator**
Hydraulic system	**NOTE – one engine-driven hydraulic pump remains; gear operation is normal but may be slower**
Operating engine	**MONITOR – power as required (39 in Hg max), cowl flap as required, mixture adjusted, TIT in limits**
Fuel crossfeed	**AS REQUIRED to use fuel from the dead-engine wing (level flight only)**
Trim	**ADJUST for straight flight**
ATC	**ADVISE – nearest suitable airport**
Passengers	**BRIEF – belts fastened, expect single-engine landing**

Air Start (Unfeather)

Restart attempt with altitude and cause identified

Fuel selector	**ON (own tank)**
Auxiliary fuel pump	**LOW**
Throttle	**1/4 OPEN**
Mixture	**FULL RICH**
Magneto switches	**ON (both)**
Propeller	**FORWARD of feather detent to mid-range (unfeathering accumulator, if fitted, spins the prop)**
Starter	**ENGAGE until propeller windmills (if no accumulator)**

▶ Engine restarted?

Yes:

Throttle	**1500 RPM until oil pressure / temperature normal, then advance SLOWLY as required**
Alternator switch	**ON**
Auxiliary fuel pump	**OFF**
Mixture / cowl flap	**AS REQUIRED**

Deferred: Engine restarted in flight – land as soon as practical, cause investigation required. **NOTED**

No:

Propeller	**FEATHER**
Mixture	**IDLE CUTOFF**

Complete Engine Securing.

Single-Engine Approach, Landing & Go-Around

One engine secured

Operating engine fuel selector	**OWN TANK (crossfeed as required, back to own tank for landing)**
Operating engine auxiliary fuel pump	**LOW**
Mixture	**FULL RICH**
Propeller	**HIGH RPM**
Cowl flap	**AS REQUIRED**
Airspeed	**101 KIAS (Vyse) until landing assured**
Approach	**NORMAL glidepath or slightly high – avoid low, flat approaches**
Landing gear	**DOWN when landing assured (within gliding distance, below 180 KIAS) – three green**
Flaps	**15° when landing assured; FULL on short final only**
Airspeed	**REDUCE below 101 KIAS only on short final (approx 95 KIAS)**

▶ Go-around required?

No:

Landing	**NORMAL – brakes as required**

Yes – above decision height:

Operating engine	**FULL POWER (advance smoothly, 39 in Hg max)**
Flaps	**UP**
Landing gear	**UP (positive climb)**
Airspeed	**101 KIAS (Vyse)**
Bank	**3–5° toward operating engine**

A single-engine go-around below approx 50 ft AGL, or once full flaps are extended, should not be attempted.

Engine Fire in Flight

LAND IMMEDIATELY

Affected engine fuel selector (memory)	**OFF**
Affected engine mixture (memory)	**IDLE CUTOFF**
Affected engine propeller (memory)	**FEATHER**
Affected engine auxiliary fuel pump (memory)	**OFF**
Affected engine throttle	**CLOSED**
Affected engine magneto switches	**OFF**
Affected engine alternator switch	**OFF**
Affected engine cowl flap	**CLOSED**
Cabin heater	**OFF**
Airspeed	**101 KIAS or higher to help extinguish fire**
Landing	**NEAREST SUITABLE AIRPORT – do not attempt restart**
Passengers	**BRIEF for emergency landing and evacuation via airstair / over-wing exits**

Engine Fire During Start

Starter (memory)	**CONTINUE CRANKING**

▶ Engine starts?

Yes:

Throttle	**1700 RPM for a few minutes**
Engine	**SHUT DOWN and inspect for damage**

No:

Mixture	**IDLE CUTOFF**
Throttle	**FULL OPEN**
Auxiliary fuel pump	**OFF**
Fuel selector	**OFF**
Starter	**CONTINUE CRANKING to draw fire into engine**
Magnetos / battery / alternators	**OFF**
Fire extinguisher	**OBTAIN / USE**
Aircraft	**EVACUATE passengers via airstair door**

Cabin / Electrical Fire

LAND AS SOON AS POSSIBLE

Battery & alternator switches (memory)	**OFF**
Avionics power switch (memory)	**OFF**
Cabin heater (memory)	**OFF**
Cabin vents	**CLOSED**
Fire extinguisher	**ACTIVATE (then ventilate cabin)**
Passengers	**BRIEF – smoke, keep low, belts fastened**

▶ Fire out and electrical power required?

Yes:

Circuit breakers	**CHECK – do not reset faulted**
Battery switch	**ON**
Alternators / avionics / equipment	**ON one at a time, check for smoke**
Deferred: Electrical fire – suspect circuits isolated, affected systems unavailable.	**NOTED**

No:

Flaps are electric and gear position lights need electrical power: plan a flaps-up landing and a hand-pump gear extension if the gear does not lock down.

Deferred: Electrical power off – flaps-up landing, no gear lights (hand pump to confirm down), no landing lights.	**NOTED**

Emergency Descent

Throttles (memory)	**CLOSED (smoothly)**
Propellers (memory)	**HIGH RPM**
Landing gear (memory)	**DOWN (below 180 KIAS)**
Flaps (memory)	**15° (below 180 KIAS)**
Airspeed (memory)	**180 KIAS**
Mixtures	**ENRICH as required**
Cowl flaps	**CLOSED**

Landing Gear Emergency Extension (Hand Pump)

Gear does not extend hydraulically

Airspeed	**BELOW 148 KIAS (slower reduces pumping effort)**
Landing gear handle	**DOWN**
Emergency hand pump	**EXTEND handle (floor between front seats) and PUMP until firm resistance**
Landing gear lights	**THREE GREEN, unlocked light out (may take 30–60 strokes; verify count in POH)**
Hand pump handle	**STOW**

Do not move the gear handle to UP after a hand-pump extension. Leave the gear down.

▶ Three green obtained?

Yes:

Deferred: Gear extended by hand pump – do not retract; hydraulic system inspection required.	**NOTED**

No:

Continue to Gear-Up / Partial Gear Landing procedure.

Gear-Up / Partial Gear Landing

Gear cannot be confirmed down

Runway	**LONGEST HARD SURFACE available, emergency services advised**
Passenger briefing	**COMPLETE (belts tight, brace, exits, do not open airstair until stopped)**
Landing gear	**UP (or as it is – do not cycle)**
Flaps	**FULL**
Approach	**NORMAL, approx 95 KIAS**
Auxiliary fuel pumps	**OFF**

On short final:

Mixtures	**IDLE CUTOFF**
Fuel selectors	**OFF**
Magneto switches	**OFF**
Battery & alternator switches	**OFF**
Touchdown	**MINIMUM SPEED, slightly nose high, wings level**
Aircraft	**EVACUATE once stopped (airstair door and over-wing exits)**

Inadvertent Spin

Intentional spins prohibited – most likely during a single-engine stall or Vmca demonstration

Throttles (memory)	**CLOSED (both)**
Rudder (memory)	**FULL OPPOSITE to rotation**
Ailerons (memory)	**NEUTRAL**
Control wheel (memory)	**BRISKLY FORWARD**
Rudder	**NEUTRALISE when rotation stops**
Recovery	**SMOOTH pull-out from dive, power as required (advance throttles slowly)**

Spin recovery in the 402C is unproven; prevention is the only sure recovery. Practise single-engine work and Vmca demonstrations at a safe altitude only.

Prototype content. Verify against the operator's approved checklist and the POH/AFM before any use. AeroE Checklist version 1.7 (build 0d080c7b) · 2026-11-07